DATE:12-08-2026

EXPERIMENT 15

Decision Tree

Aim

To implement a **Decision Tree classification algorithm** using Python.

Algorithm

1. Store the training data.
2. Calculate the best feature for splitting the data.
3. Split the data into branches.
4. Repeat the process until a decision can be made.
5. Give a new input to the decision tree.
6. Predict the output class.
7. Display the result.

Program

```
# Decision Tree without sklearn

# Training data
data = [
    [2, 60, "Fail"],
    [3, 65, "Fail"],
    [4, 70, "Pass"],
    [5, 75, "Pass"],
    [6, 80, "Pass"],
    [7, 85, "Pass"]
]


# Decision Tree prediction function
def decision_tree(study_hours, attendance):

    if study_hours <= 3:
        return "Fail"

    elif attendance < 70:
        return "Fail"

    else:
        return "Pass"
```

```
# Display Decision Tree
def display_tree():
    print("\nDecision Tree:\n")

    print("                  Study Hours")
    print("                    <= 3 ?")
    print("                   /       \\")
    print("                 Yes        No")
    print("                 /            \\")
    print("              FAIL          Attendance")
    print("                              < 70 ?")
    print("                             /     \\")
    print("                           Yes      No")
    print("                           /         \\")
    print("                        FAIL        PASS")


# Input
study_hours = int(input("Enter study hours: "))
attendance = int(input("Enter attendance percentage: "))


# Prediction
result = decision_tree(study_hours, attendance)


# Output
print("\nStudy Hours:", study_hours)
print("Attendance:", attendance)
print("Predicted Result:", result)

display_tree()
```

Input

Enter study hours: 5

Enter attendance percentage: 75

Output

```
Enter study hours: 5
Enter attendance percentage: 75

Study Hours: 5
Attendance: 75
Predicted Result: Pass

Decision Tree:

                    Study Hours
                       <= 3 ?
                     /        \
                   Yes         No
                   /             \
                FAIL          Attendance
                                < 70 ?
                               /      \
                           Yes         No
                           /             \
                        FAIL           PASS
```

Result

Thus, the **Decision Tree classification algorithm** was successfully implemented in Python without using external libraries.